Domain Adaptive for Detection

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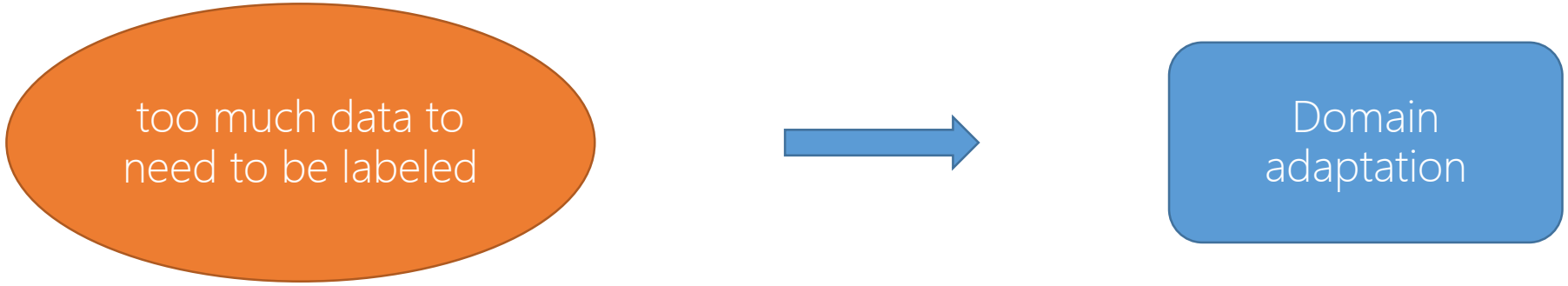
Domain adaptation (Ben-David et al., 2006, 2010), that suggests that a good representation for cross-domain transfer is one for which an algorithm cannot learn to identify the domain of origin of the input observation.



Task description

- Source data: (x^s, y^s) → Training data
- Target data: (x^t) → Testing data

Same task, mismatch

MNIST

SOURCE

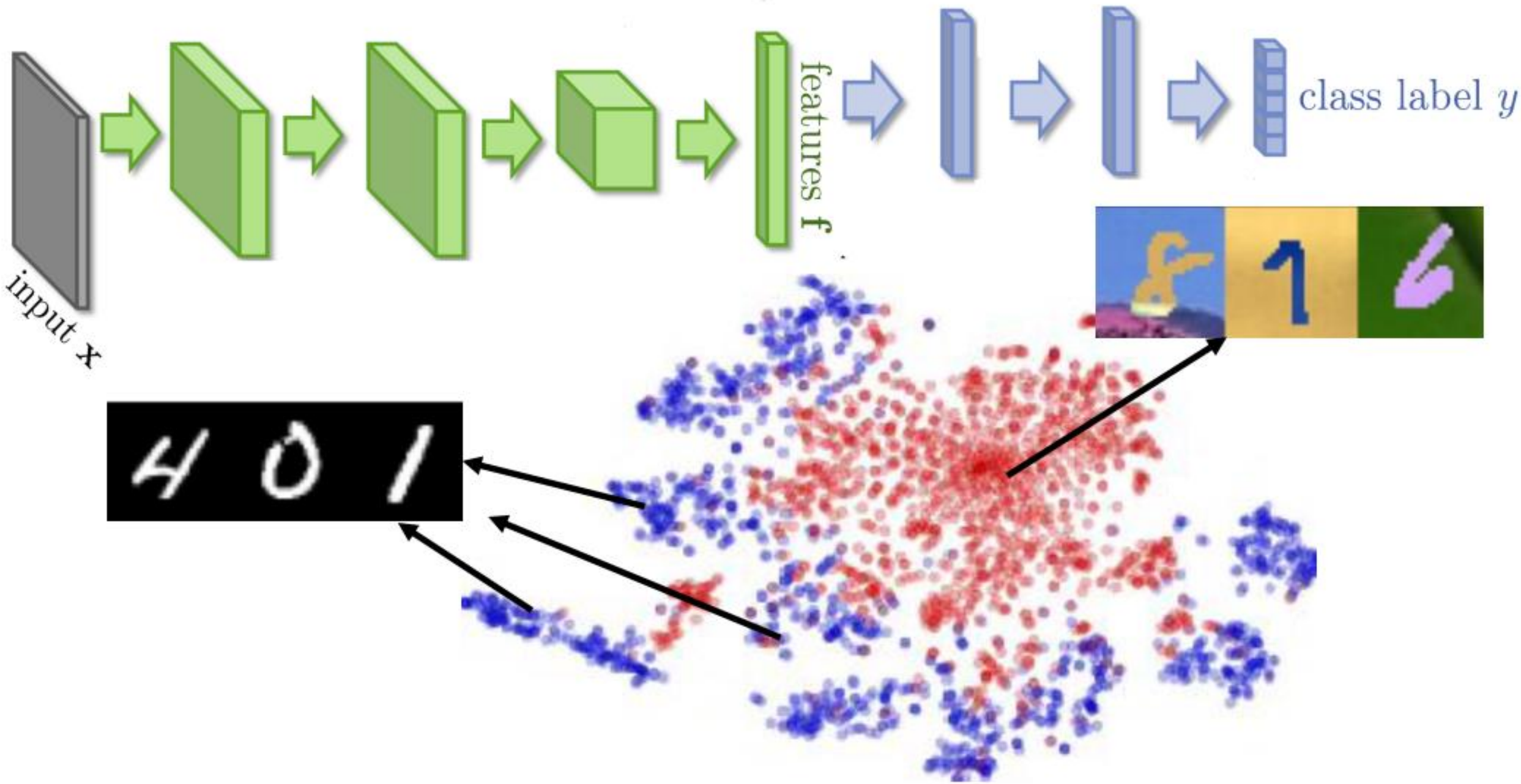
with label

TARGET

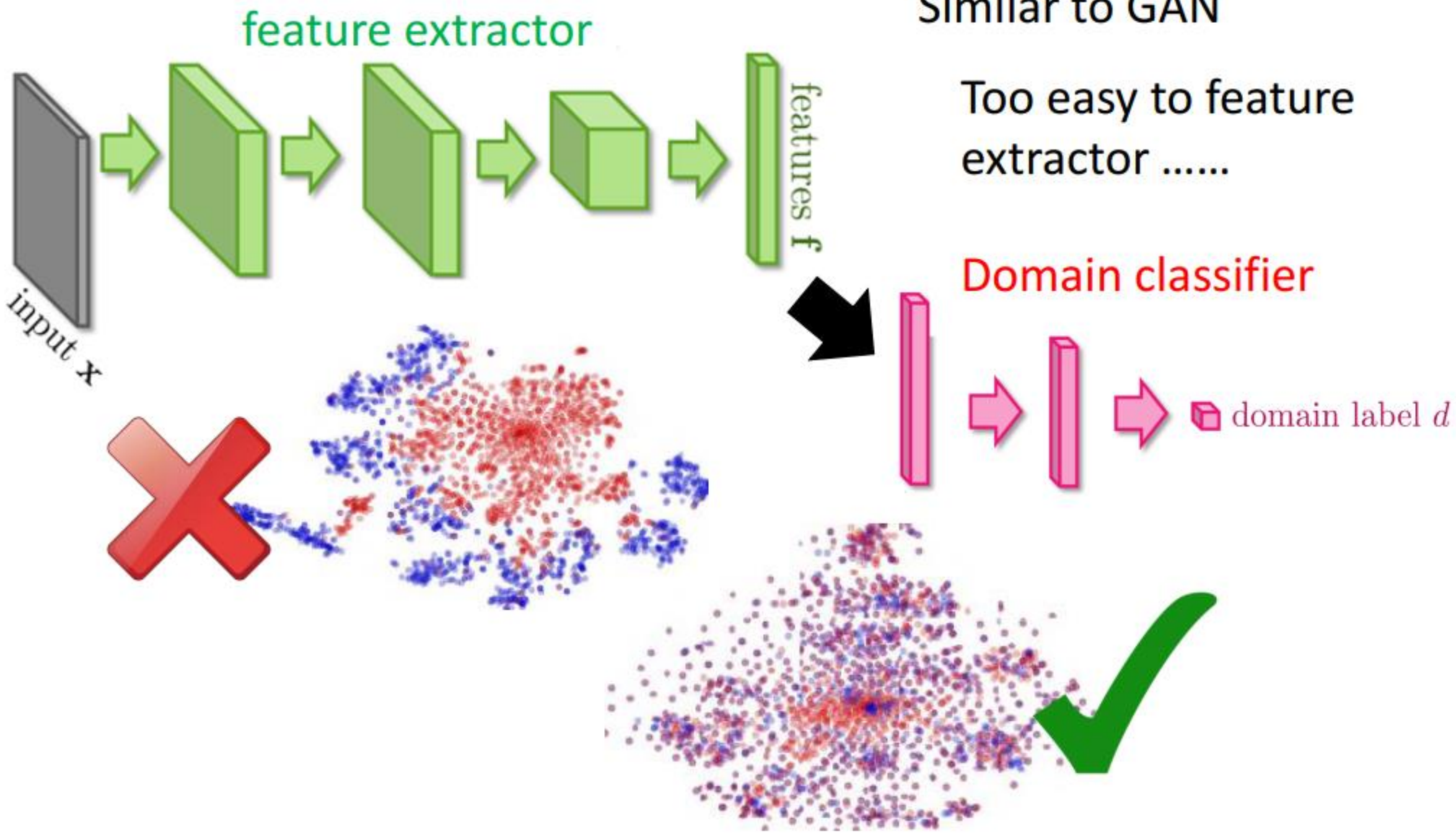
without label

MNIST-M

Domain-adversarial training



Domain-adversarial training



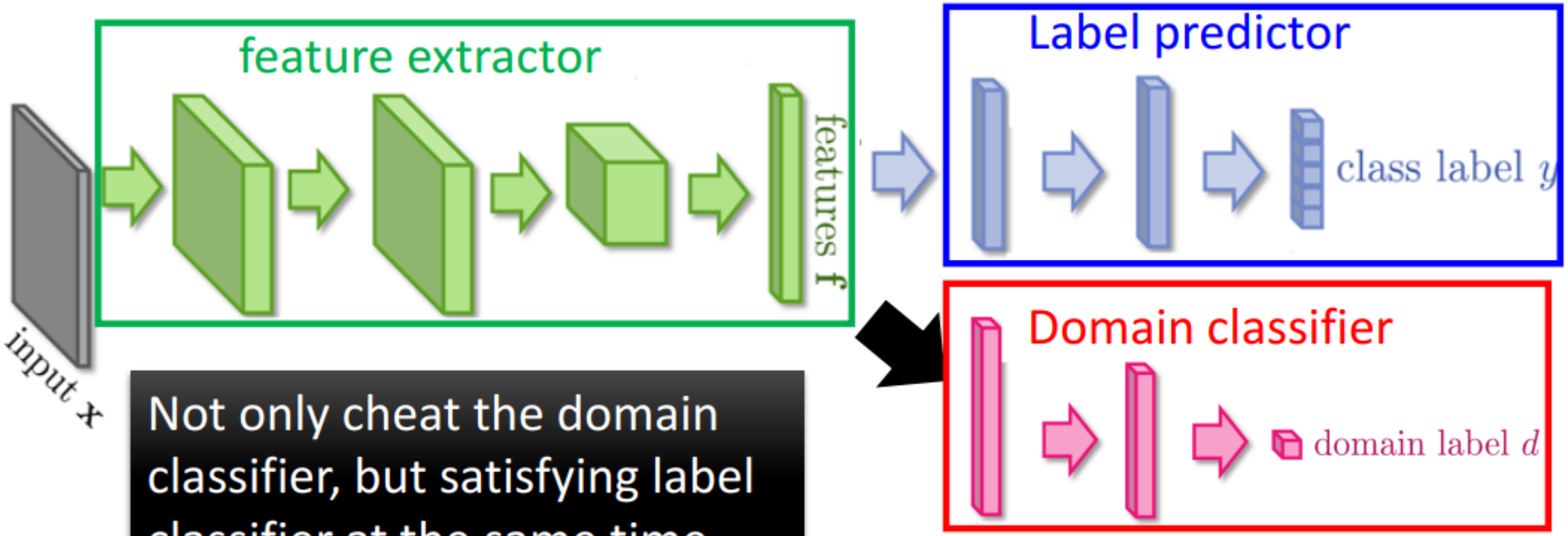
Similar to GAN

Too easy to feature extractor ……

Domain-adversarial training

Maximize label classification accuracy + minimize domain classification accuracy

Maximize label classification accuracy

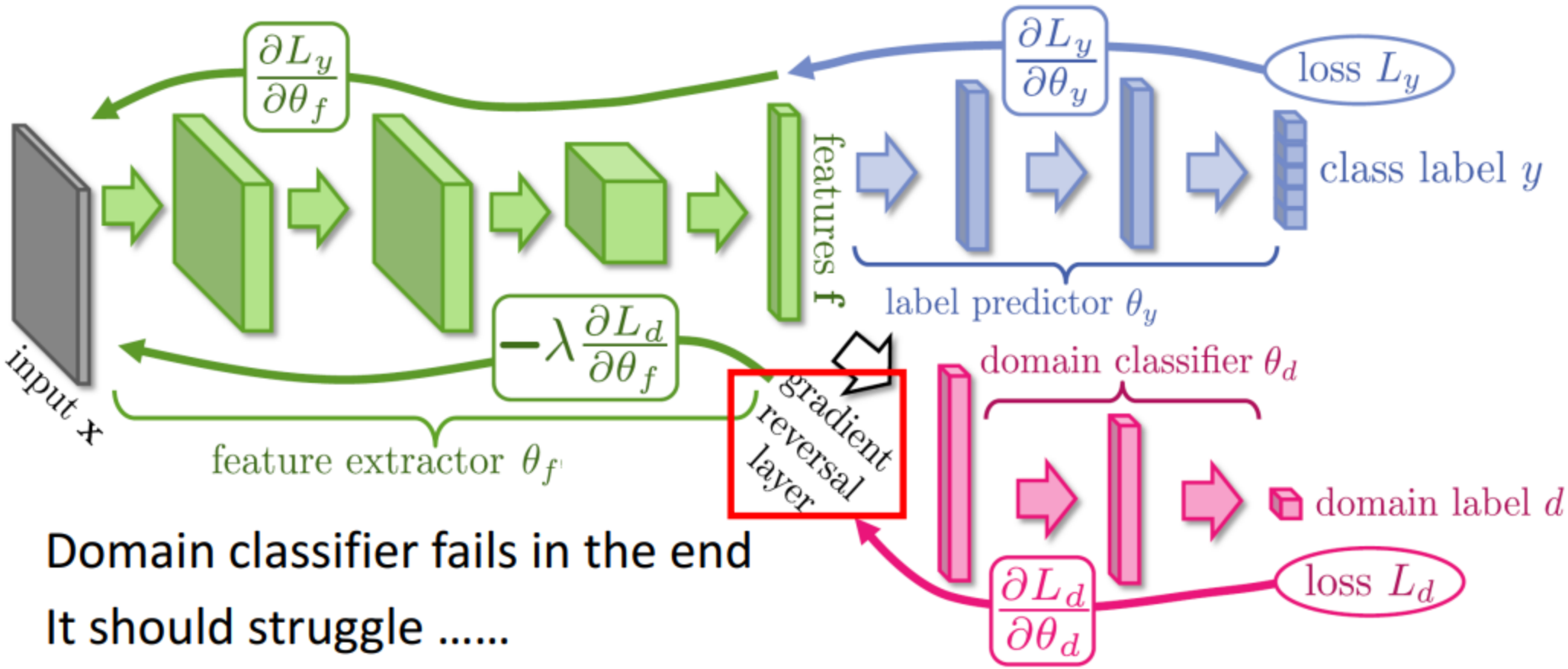


Not only cheat the domain classifier, but satisfying label classifier at the same time

Maximize domain classification accuracy

This is a big network, but different parts have different goals.

Domain-adversarial training



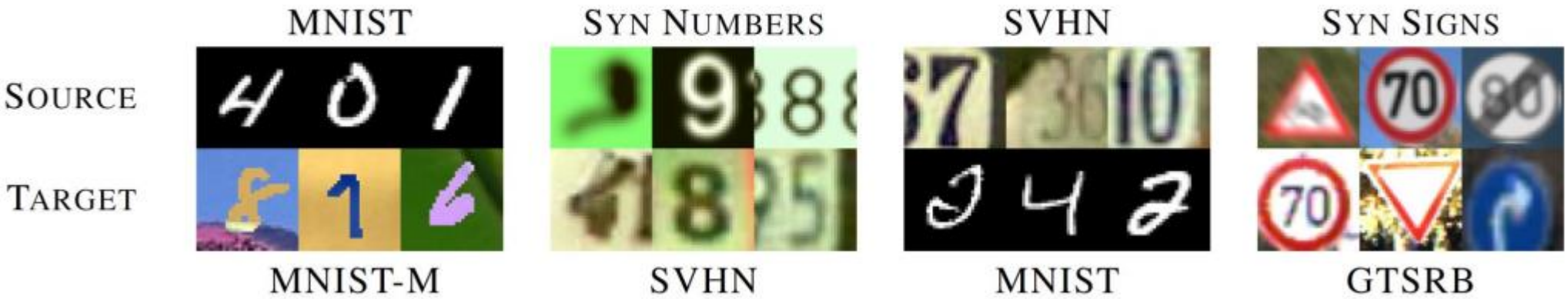
Domain classifier fails in the end

It should struggle ……

Yaroslav Ganin, Victor Lempitsky, Unsupervised Domain Adaptation by Backpropagation, ICML, 2015

Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, Domain-Adversarial Training of Neural Networks, JMLR, 2016

Domain-adversarial training



METHOD	SOURCE	MNIST	SYN NUMBERS	SVHN	SYN SIGNS
	TARGET	MNIST-M	SVHN	MNIST	GTSRB
SOURCE ONLY		.5749	.8665	.5919	.7400
SA (FERNANDO ET AL., 2013)		.6078 (7.9%)	.8672 (1.3%)	.6157 (5.9%)	.7635 (9.1%)
PROPOSED APPROACH		**.8149** (57.9%)	**.9048** (66.1%)	**.7107** (29.3%)	**.8866** (56.7%)
TRAIN ON TARGET		.9891	.9244	.9951	.9987

Yaroslav Ganin, Victor Lempitsky, Unsupervised Domain Adaptation by Backpropagation, ICML, 2015

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Figure 1. **Illustration of different datasets for autonomous driving:** From top to bottom-right, example images are taken from: *KITTI*[17], *Cityscapes*[5], *Foggy Cityscapes*[49], *SIM10K*[30]. Though all datasets cover urban scenes, images in those dataset vary in style, resolution, illumination, object size, *etc*. The visual difference between those datasets presents a challenge for applying an object detection model learned from one domain to another domain.

The reason of the domain shift:

(1) image level :
scale, style, illumination
(2) instance level:
object appearance, size and orientation, etc.

Chen Y, Li W, Sakaridis C, et al. Domain Adaptive Faster R-CNN for Object Detection in the Wild[J]. 2018.

Warnings:

Something of the **Probabilistic Perspective** for DA Faster Rcnn:

Given some definition:

The object detection problem can be viewed as learning the posterior $P(C,B|I)$, where I is the image representation, B is the bounding-box of an object and C *is* the category of the object.

We use $Ps(C, B, I)$ and $Pt(C, B, I)$ to denote the source domain joint distribution and the target domain joint distribution.

We know that: $Ps(C, B, I) \neq Pt(C, B, I)$

Chen Y, Li W, Sakaridis C, et al. Domain Adaptive Faster R-CNN for Object Detection in the Wild[J]. 2018.

Image-Level Adaptation:

$P(C, B, I) = P(C, B| I) * P(I)$ 我们假设在两个domain中的$P(C, B| I)$是相同的，因此造成domain shift的原因是$P(I)$。

$$\boxed{Ps(I) = Pt(I)}$$

Chen Y, Li W, Sakaridis C, et al. Domain Adaptive Faster R-CNN for Object Detection in the Wild[J]. 2018.

Instance-Level Adaptation:

$P(C, B, I) = P(C \mid B, I) * P(B, I)$ 我们假设在两个domain中的$P(C \mid B, I)$是相同的，因此造成domain distribution shift的原因是$P(B, I)$。

$$\boxed{Ps(B, I) = Pt(B, I)}$$

Chen Y, Li W, Sakaridis C, et al. Domain Adaptive Faster R-CNN for Object Detection in the Wild[J]. 2018.

Joint Adaptation:

$P(B, I) = P(B \mid I) * P(I)$ and the condition distribution $P(B \mid I)$ is assumed to be the same and non-zero for domains, thus we have:

$$Ps(I) = Pt(I) \iff Ps(B, I) = Pt(B, I)$$

To make the equation achieved we must fix two problems:

$P(I)$ and $P(B \mid I)$ ← **$P(B \mid I)$** is learned using the source domain data only, which is easily biased toward the source domain

Chen Y, Li W, Sakaridis C, et al. Domain Adaptive Faster R-CNN for Object Detection in the Wild[J]. 2018.

By denoting the domain label as D, the image-level domain classifier can be view as estimating P(D | I), and the instance-level domain classifier can be seen as estimating P(D | B, I).

$$P(D \mid B, I) * P(B \mid I) = P(B \mid D, I) * P(D \mid I)$$

$$P(D \mid I) = P(D \mid B, I) \quad \Rightarrow \quad P(B \mid I) = P(B \mid D, I)$$

Chen Y, Li W, Sakaridis C, et al. Domain Adaptive Faster R-CNN for Object Detection in the Wild[J]. 2018.

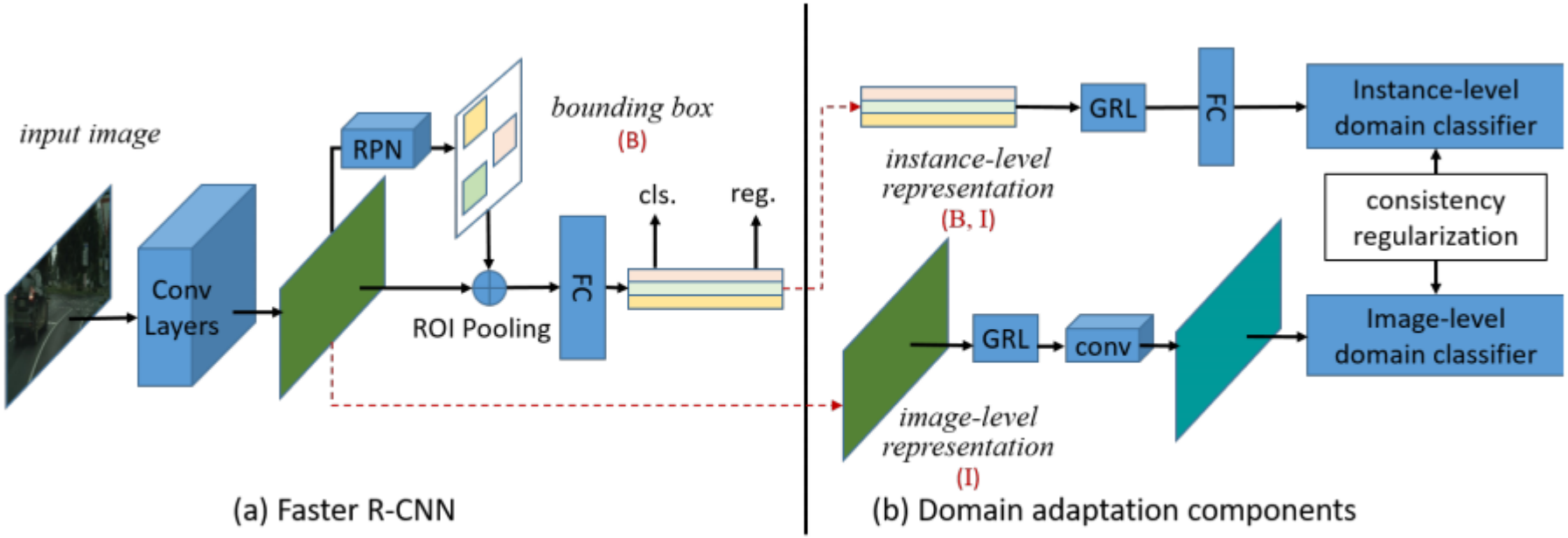

Figure 2. **An overview of our Domain Adaptive Faster R-CNN model:** we tackle the domain shift on two levels, the image level and the instance level. A domain classifier is built on each level, trained in an adversarial training manner. A consistency regularizer is incorporated within these two classifiers to learn a domain-invariant RPN for the Faster R-CNN model.

Chen Y, Li W, Sakaridis C, et al. Domain Adaptive Faster R-CNN for Object Detection in the Wild[J]. 2018.

	img	ins	cons	person	rider	car	truck	bus	train	mcycle	bicycle	mAP
Faster R-CNN				17.8	23.6	27.1	11.9	23.8	9.1	14.4	22.8	18.8
Ours	✓			22.9	30.7	39.0	20.1	27.5	17.7	21.4	25.9	25.7
		✓		23.6	30.6	38.6	20.8	**40.5**	12.8	17.1	26.1	26.3
	✓	✓		24.2	**31.2**	39.1	19.1	36.2	19.2	17.1	27.0	26.6
	✓	✓	✓	**25.0**	31.0	**40.5**	**22.1**	35.3	**20.2**	**20.0**	**27.1**	**27.6**

Table 2. Quantitative results on the *Foggy Cityscapes* validation set, models are trained on the *Cityscapes* training set.

	img	ins	cons	K → C	C → K
Faster R-CNN				30.2	53.5
Ours	✓			36.6	60.9
		✓		34.6	57.6
	✓	✓		37.3	62.7
	✓	✓	✓	**38.5**	**64.1**

Table 3. Quantitative analysis of adaptation result between *KITTI* and *Cityscapes* . We report AP of *Car* on both directions. *e.g.* K → C and C → K.

	img	ins	cons	car AP
Faster R-CNN				30.12
Ours	✓			33.03
		✓		35.79
	✓	✓		37.86
	✓	✓	✓	**38.97**

Table 1. The average precision (AP) of *Car* on the *Cityscapes* validation set. The models are trained using the *SIM 10k* dataset as the source domain and the *Cityscapes* training set as the target domain. *img* is short for *image-level alignment*, *ins* for *instance-level alignment* and *cons* is short for our *consistency loss*

Chen Y, Li W, Sakaridis C, et al. Domain Adaptive Faster R-CNN for Object Detection in the Wild[J]. 2018.

Transfer Learning - Overview

		Source Data (not directly related to the task)	
		labelled	unlabeled
Target Data	labelled	Fine-tuning Multitask Learning	Self-taught learning Rajat Raina , Alexis Battle , Honglak Lee , Benjamin Packer , Andrew Y. Ng, Self-taught learning: transfer learning from unlabeled data, ICML, 2007
	unlabeled	Domain-adversarial training Zero-shot learning	Self-taught Clustering Wenyuan Dai, Qiang Yang, Gui-Rong Xue, Yong Yu, "Self-taught clustering", ICML 2008

Different from semi-supervised learning